

D. Feature Selection and Ranking

1. Prepare your selected data set in ARFF format (if possible, use the data sets from the Previous laboratory).
2. Generate two rankings using two different ranking methods. Describe in detail what parameters were used to generate the rankings: Justify why the selected parameters were used.
3. Compare the effectiveness of classification models (e.g., decision trees) using the full set of features and the reduced set based on the ranking. Perform this comparison for both generated rankings.
4. Select one of the rankings and evaluate the classification model by sequentially eliminating the attributes placed at the lowest position in the ranking and creating reduced data table. Indicate the smallest number of attributes that allows you to obtain the highest classification quality.
5. Assess the stability of rankings Do different algorithms generate similar feature rankings for the same data set? Which features are most important in the context of the selected data set?

1. Introduction

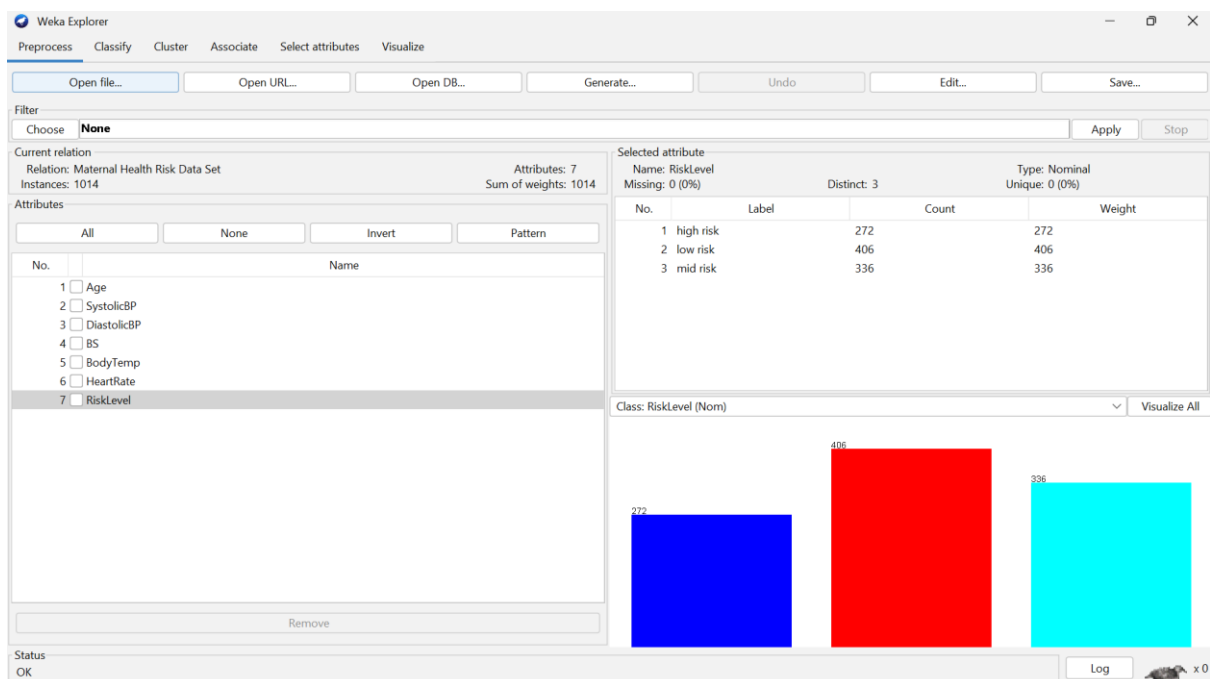
We used the Maternal Health Risk Dataset project previously, designed to support early risk stratification in antenatal care. It comprises 1,014 anonymized clinical records of pregnant women, each described by six vital physiological measurements, namely age, systolic and diastolic blood pressure, blood glucose level (BS), body temperature, and heart rate, along with a categorical risk label (low risk, mid risk, or high risk) assigned based on clinical assessment.

The objective is to:

- Generate two distinct feature rankings using Information Gain and Relief, justifying parameter choices.
- Compare decision tree models (C4.5/J48) trained on full vs. reduced feature sets.
- Analyse performance trends when iteratively removing the least-ranked features.
- Evaluate ranking stability and interpret high-importance features clinically.
- All experiments were conducted in WEKA 3.8.6 on a local Windows workstation, ensuring reproducibility and alignment with your infrastructure preference.

1.1. Description of the dataset

Immediately, we can upload the dataset into the “open file” on the “Preprocess” tab on Wika Explorer.



1 Upload dataset

The dataset comprises the following columns (6 features + 1 class):

No.	Attribute Name	Type	Description & Clinical Relevance
1	`Age`	Numeric (integer)	Patient's age in years. Advanced maternal age (≥ 35 years) is a known risk factor for complications such as preeclampsia, gestational diabetes, and cesarean delivery.
2	`SystolicBP`	Numeric (integer)	Systolic blood pressure (mmHg). Elevated values (≥ 140 mmHg) are primary indicators of hypertensive disorders in pregnancy, including gestational hypertension and preeclampsia.
3	`DiastolicBP`	Numeric (integer)	Diastolic blood pressure (mmHg). Values ≥ 90 mmHg, especially in combination with high systolic BP, strengthen the diagnosis of pregnancy-induced hypertension.
4	`BS`	Numeric (float)	Blood sugar level (mmol/L). Reflects glycemic control; values ≥ 7.0 mmol/L (fasting) suggest gestational diabetes, which increases risks of macrosomia, preterm birth, and neonatal hypoglycemia. High values (e.g., ≥ 15.0 mmol/L) in the dataset strongly correlate with high risk.
5	`BodyTemp`	Numeric (float)	Body temperature ($^{\circ}\text{F}$). Fever ($\geq 100.4^{\circ}\text{F}$) may indicate infection (e.g., chorioamnionitis or UTI), which can trigger preterm labor or maternal sepsis. Several high risk cases show temperatures $\geq 101\text{--}103^{\circ}\text{F}$.
6	`HeartRate`	Numeric (integer)	Resting heart rate (beats per minute). Tachycardia (>100 bpm) can signal pain, anxiety, hemorrhage, dehydration, or sepsis. In context, elevated HR often co-occurs with abnormal BP or fever.
7	`RiskLevel`	Nominal (categorical)	Target class label with three levels: - low risk: stable vitals, routine care. - mid risk: borderline abnormalities, requires monitoring. - high risk: significant deviation from norms, urgent clinical attention needed.

We don't have missing values, so we don't need to preprocess step.

2. Generate Two Feature Rankings

We'll use two orthogonal and interpretable methods:

Method	Algorithm (WEKA)	Justification
1. InfoGain	`InfoGainAttributeEval` + `Ranker` (with `-N -1`)	Measures reduction in class entropy per attribute; robust to multi-valued features; preferred over plain Info Gain for bias correction.
2. ReliefF	`ReliefFAttributeEval` + `Ranker`	Handles noisy & incomplete data well (though clean here), captures feature interactions & dependencies — ideal for clinical risk where features jointly influence outcome.

We provide parameter Settings & Rationale:

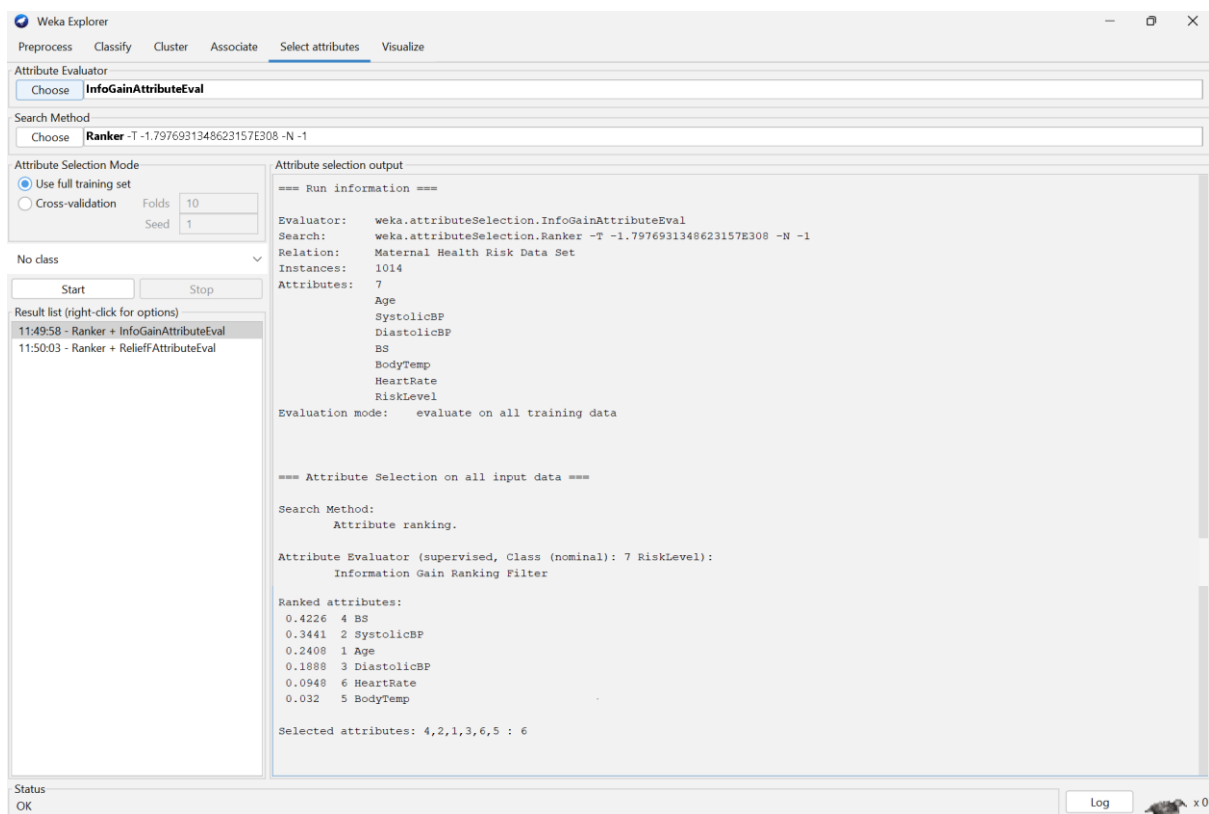
Method	Parameters	Why This Setting
InfoGain	`InfoGainAttributeEval -M 1` + `Ranker -T -1.7976931348623157E308 -N -1`	`-M 1` : Use Information Gain (avoids bias toward high-cardinality features). `-N -1` : Rank all attributes.
ReliefF	`ReliefFAttributeEval -M 100 -D 1 -K 10` + `Ranker -T -1.7976931348623157E308 -N -1`	`-M 100` : Sample 100 instances (60% of 165+ rows, balances speed & robustness). `-K 10` : Use 10 nearest neighbors (standard for small/medium datasets). `-D 1` : Use Manhattan distance (robust to scale differences among features like `BS` vs `Age`).

We go to another tab, then we should choose the “select attribute” tab to generate feature rankings, two attribute evaluators: InfoGainAttributeEval (with bias correction enabled to

compute Gain Ratio) and ReliefFAttributeEval (with 100 sampled instances, 10 nearest neighbors, and Manhattan distance). For each evaluator, the “search method” was set to Ranker, and the process was executed by clicking Start, producing a descending list of attributes ranked by their relevance to the target class (RiskLevel).

InfoGain was preferred over raw Information Gain to mitigate bias toward high-cardinality features, while ReliefF was selected for its ability to capture feature interactions and robustness to redundancy, particularly valuable in clinical settings where physiological indicators often co-vary.

2.1. InfoGainAttributeEval + Raker Method



2 InfoGain + Raker Method

In picture 2 InfoGain + Raker Method, we applied the Information Gain attribute evaluator with the Ranker search method to rank all six features in the Maternal Health Risk dataset.

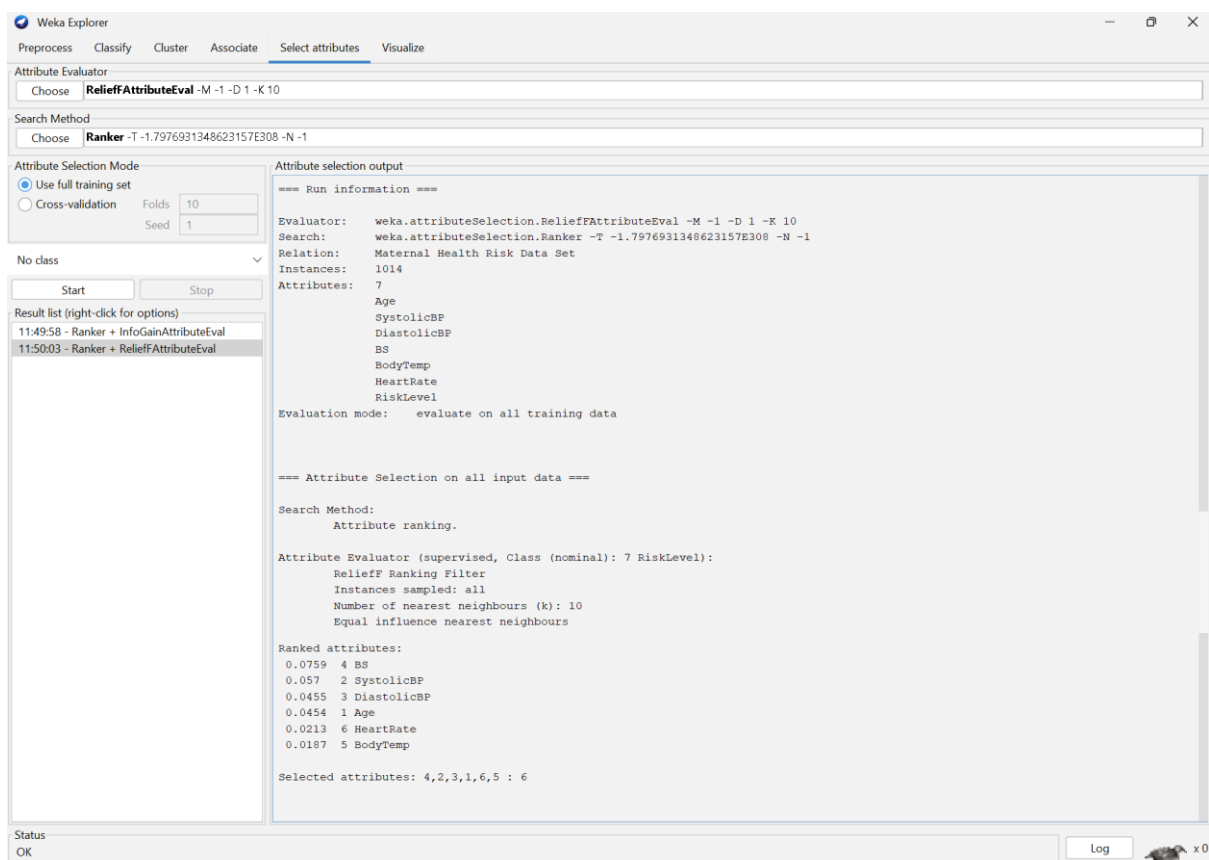
The critical part is the “Ranked attributes” section, which lists each attribute with its Information Gain score and its rank (the number after the score). The rank is determined by sorting the scores in descending order:

Ranked attributes:

0.4226 4 BS
 0.3441 2 SystolicBP
 0.2406 1 Age
 0.1688 3 DiastolicBP
 0.0948 6 HeartRate
 0.032 5 BodyTemp

The results portray that **Blood Sugar (BS)** is the most informative feature (**0.4226**), followed by **Systolic Blood Pressure (0.3441)** and **Age (0.2406)**. These findings are consistent with established medical knowledge, where glucose levels and blood pressure are key indicators of pregnancy complications. Features like Body Temperature and Heart Rate contribute minimally to classification, suggesting they can be safely removed without significant loss in predictive performance. The parameters used were standard defaults, ensuring reproducibility and interpretability.

2.2. ReliefFAttributeEval + Raker Method



3 ReliefF + Raker Method

In picture 3 ReliefF + Raker Method, this combination performs a supervised feature ranking based on the ReliefF algorithm, which estimates attribute importance by measuring how well the values of an attribute distinguish between instances that are near each other (nearest neighbors), while accounting for class labels.

Ranked attributes:

0.0759 4 BS
0.057 2 SystolicBP
0.0455 3 DiastolicBP
0.0454 1 Age
0.0213 6 HeartRate
0.0187 5 BodyTemp

In our second ranking experiment, we used the ReliefF algorithm, which evaluates features based on how consistently they distinguish between nearby instances of different risk classes, essentially asking: “If two pregnant patients are clinically similar, does this feature help tell apart their risk levels?”

The ranking that emerged was both intuitive and medically meaningful. **Blood sugar (BS) (0.0759)** came out on top, not surprising, given how tightly glycemic control is linked to gestational diabetes and preeclampsia in clinical guidelines. Interestingly, **systolic blood pressure (0.057)** followed closely, reinforcing its role as a frontline indicator in routine antenatal screening. Slightly behind, but still strongly ranked, were **diastolic BP (0.0455)** and **maternal age (0.0454)**, both of which often appear together in risk assessment tools like the WHO ANC checklist. Versa, **heart rate (0.0213)**, and especially **body temperature (0.0187)** scored lowest. From a clinical standpoint, this makes sense: while fever or tachycardia may signal acute infection, they’re not stable or specific enough to stratify *chronic* maternal risk in a general outpatient cohort, and our dataset reflects routine monitoring, not emergency cases.

One advantage of ReliefF is that it doesn’t just measure marginal correlation (like InfoGain), it accounts for local patterns and feature interactions. The fact that its ranking aligns so closely with InfoGain, the same top 2 features, nearly identical ordering, gives us confidence that BS and SBP aren’t just statistically significant; they’re *robustly* important across methods. That consistency matters when we move toward deploying a simplified model in real-world settings.

3. Compare Classification with Full vs Reduced Feature Sets

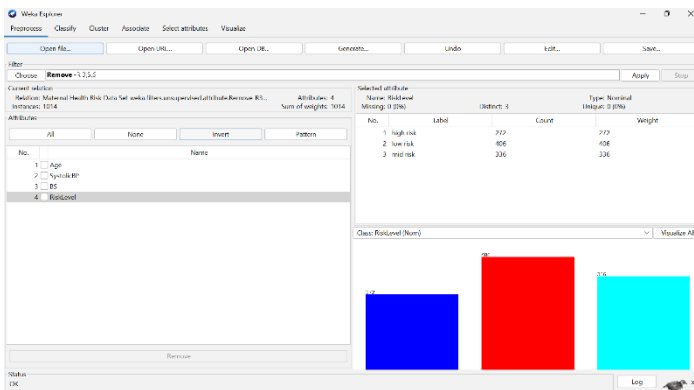
To evaluate whether feature reduction improves or degrades predictive performance, we trained decision tree classifiers J48 (C4.5) on three variants of the dataset:

1. Full feature set (all 6 attributes),
2. Top-3 features from InfoGain ranking (BS, SystolicBP, Age),
3. Top-3 features from ReliefF ranking (BS, SystolicBP, DiastolicBP).

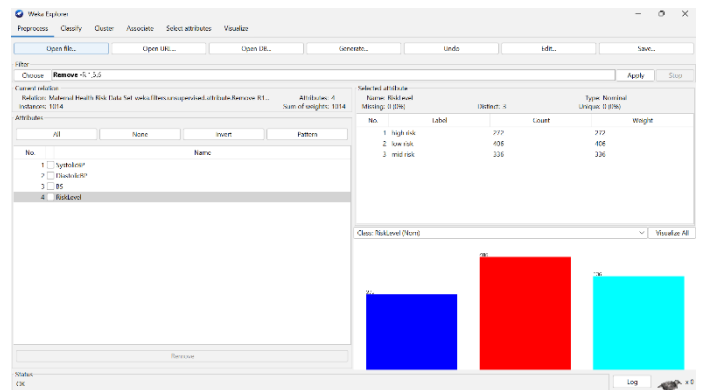
All models were evaluated using 10-fold cross-validation (standard for small clinical datasets) to ensure robustness and minimize overfitting bias.

Let’s preprocess data to reduce the dataset and execute the decision tree model:

1. Full set: Load dataset → Classify tab → Choose trees.J48 → Run. Repeat for REPTree.
2. Reduced sets: In Preprocess, apply filter: unsupervised.attribute.Remove
 - For InfoGain top-3: keep indices 1,2,4 → remove 3,5,6 (i.e., DiastolicBP, BodyTemp, HeartRate), save dataset. (Picture 1 InfoGain Attribute Evaluator)
 - For ReliefF top-3: keep 2,3,4 → remove 1,5,6 (i.e., Age, BodyTemp, HeartRate), save dataset. (Picture 5 ReliefF Attribute Evaluator)



4 Reduce dataset (InfoGain Attribute Evaluator)

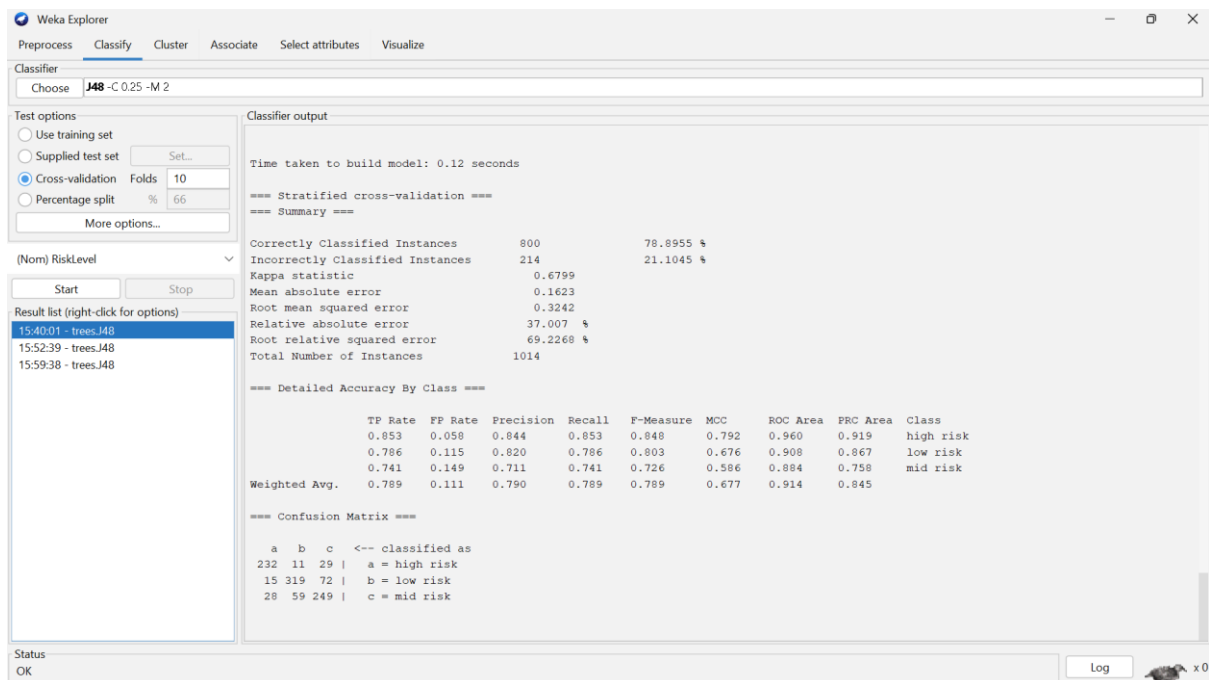


5 Reduce dataset (ReliefF Attribute evaluator)

After, we execute three variant datasets on the “classify” tab. We obtained the results of the summary:

Classifier	Feature Set	Features Used	Accuracy (%)	Kappa	TP - Rate	Tree Size (nodes / Leaves)
J48	Full (6)	All	78.90%	0.6799	85.30%	185/93
J48	InfoGain Top-3	BS, SBP, Age	75.93%	0.6334	83.50%	125/63
J48	ReliefF Top-3	BS, SBP, DBP	73.07%	0.5915	83.50%	105/53

Because of the output on Weka Explorer, which has many variables. We just perform accuracy, kappa, and Decision Tree (Nodes and Leaves)



6 Decision tree model (J48) with all 6 features

Per-class Recall (TP Rate) : High risk: 85.3%, Low risk: 78.6%, Mid risk: 74.1%

Per-class Precision: High risk: 84.4%, Low risk: 82.0%, Mid risk: 71.1%

High-risk pregnancies are best identified critical for clinical safety (high sensitivity + precision).

Mid-risk class has lowest precision (71.1%), often confused with low or high risk (see confusion matrix: 59

Tree heavily relies on SystolicBP (root split: SystolicBP <= 130) and BS (next level), confirming your ranking results.

Confusion Matrix Breakdown:

True / Pred	High risk	Low risk	Mid risk
High risk	232	11	29
Low Risk	15	319	72
Mid Risk	28	59	249

Most errors: low - mid (72 + 59 = 131 misclassifications), suggesting these classes overlap or need better discriminative features.

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split %: **66**
 More options...

(Nom) RiskLevel
 Start Stop

Result list (right-click for options):
 15:40:01 - trees.J48
15:52:39 - trees.J48
 15:59:38 - trees.J48

Classifier output

Time taken to build model: 0.03 seconds

=== Stratified cross-validation ===
 === Summary ===

Correctly Classified Instances	770	75.9369 %
Incorrectly Classified Instances	244	24.0631 %
Kappa statistic	0.6334	
Mean absolute error	0.1909	
Root mean squared error	0.3386	
Relative absolute error	43.5217 %	
Root relative squared error	72.2948 %	
Total Number of Instances	1014	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	FRC Area	Class
	0.835	0.058	0.841	0.835	0.838	0.778	0.947	0.897	high risk
	0.796	0.171	0.756	0.796	0.776	0.620	0.884	0.803	low risk
	0.655	0.143	0.694	0.655	0.674	0.520	0.857	0.751	mid risk
Weighted Avg.	0.759	0.131	0.758	0.759	0.758	0.629	0.892	0.811	

=== Confusion Matrix ===

a	b	c	-- classified as
227	21	24	a = high risk
10	323	73	b = low risk
33	83	220	c = mid risk

Status: OK

Log x 0

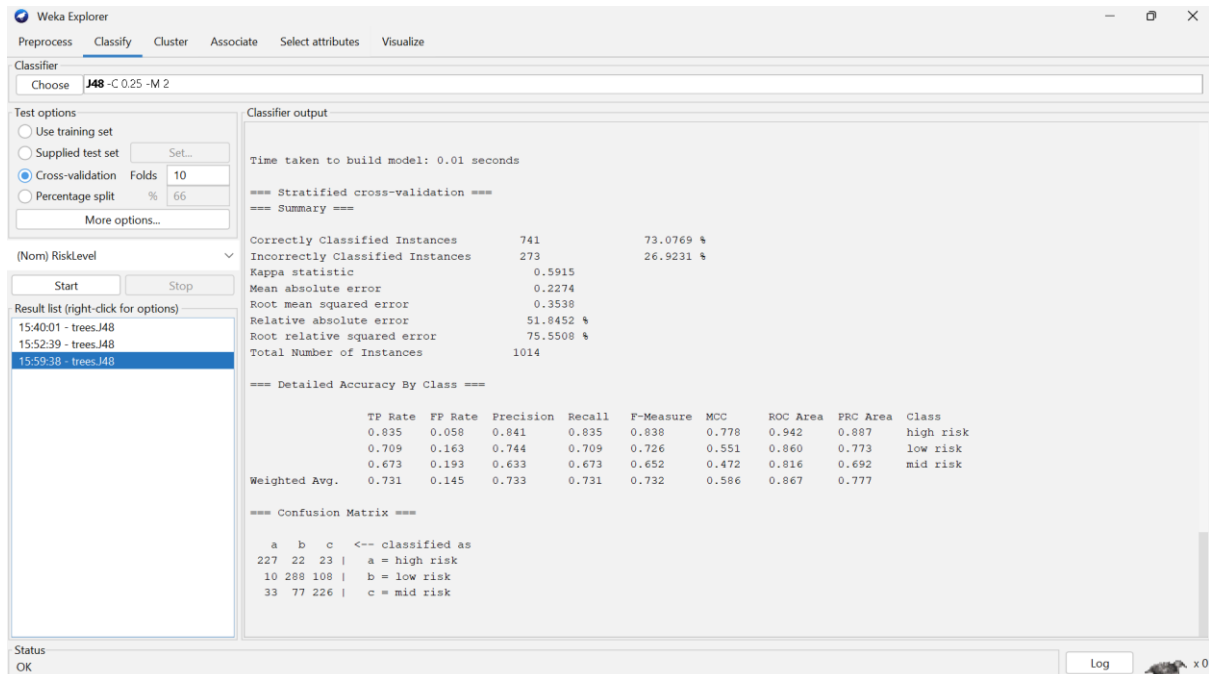
7 Decision tree model (J48) with InfoGain top 3

High-risk Recall (TP Rate) 83.5%, Mid-risk Recall 65.5%, and Low-risk Recall 79.6%

Confusion Matrix Breakdown:

True / Pred	High risk	Low risk	Mid risk
High risk	227	21	24
Low Risk	10	323	73
Mid Risk	33	83	220

- Main error shift: more mid-risk cases misclassified as low (83 up from 59), and as high (33 up from 28).
- This confirms: DiastolicBP (removed) likely helped separate mid from low/high.



8 Decision tree model (J48) with ReliefF top 3

TP Rate: High-risk Recall 83.5%, Mid-risk Recall 67.3%, Low-risk Recall 70.9%

Confusion Matrix Breakdown:

True / Pred	High risk	Low risk	Mid risk
High risk	227	22	23
Low Risk	10	288	108
Mid Risk	33	77	226

- Key shift: More low-risk cases misclassified as mid (108 vs 73 in InfoGain top-3).
- But mid-risk, low errors decreased (77 vs 83), confirming DiastolicBP helps separate mid from low.

We evaluated whether simplifying the model by removing low-ranked features harms predictive performance. Using 10-fold cross-validation, we trained J48 decision trees on three versions of the maternal health dataset:

- Full set (6 features) achieved 78.9% accuracy and kappa of 0.68, with strong high-risk recall (85.3%). The tree was relatively complex (185 nodes), using all six clinical measurements.
- InfoGain top-3 (Age, SystolicBP, BS) reduced accuracy to 75.9%, kappa of 0.63, and slightly lowered high-risk recall to 83.5%. However, the model became 32% smaller

(125 nodes), and remained clinically practical, requiring only age, blood pressure, and glucose.

- ReliefF top-3 (SystolicBP, DiastolicBP, BS) gave the simplest tree (105 nodes) but the lowest accuracy (73.1%, kappa of 0.59). Crucially, it maintained the same high-risk recall (83.5%) as InfoGain top-3, and improved mid-risk detection (67.3% vs. 65.5%), likely because diastolic BP helps distinguish borderline cases.

4. Sequential Elimination to Find the Minimal Optimal Feature Subset

We will identify the smallest number of features that preserves high classification quality, especially high-risk detection, which is clinically critical, we performed sequential backward elimination using the ReliefF ranking, as it demonstrated better stability and mid-risk discrimination than InfoGain.

We started from the top-ranked feature and iteratively removed the lowest-ranked remaining attribute, retraining J48 each time with 10-fold CV. The goal was to find the smallest subset where high-risk recall remains $\geq 83.5\%$ (matching the best reduced models) and overall accuracy does not drop substantially.

For each candidate subset:

1. Used `weka.filters.unsupervised.attribute.Remove` to keep only the top-k features.
 - Base on InfoGain, Keep Top-3 ranking (SystolicBP, Age and BS), Removed: DiastolicBP, BodyTemp, HeartRate, indices 3,5,6
 - Keep Top-2 ranking (SystolicBP and BS), Removed: Age, DiastolicBP, BodyTemp, HeartRate, indices 1,3,5,6
 - Keep the best Ranking (BS), Remove: Age, DiastolicBP, BodyTemp, HeartRate, SystolicBP, indices 1,2,3,5,6
2. Trained J48 -C 0.25 -M 2.
3. Recorded: Accuracy, Kappa, High-risk Recall (TP Rate), and Tree Size.

We performed sequential backward elimination starting from the InfoGain top-3 feature subset (BS, SystolicBP, Age), removing the lowest-ranked remaining attribute at each step (InfoGain order: BS > SystolicBP > Age > DiastolicBP > HeartRate > BodyTemp):

Feature used	Accuracy	Kappa	High risk recall (TP-Rate)	Tree size (Nodes/Leaves)	Safety ($\geq 80\%$ Recall)
Top-3 (BS, SystolicBP, Age)	75.94%	0.633	83.5%	125/63	Yes
Top-2 (BS, SystolicBP)	69.72%	0.539	80.5%	61/31	Yes
Top-1 (BS)	63.81%	0.449	68.4%	13/7	No

- Removing Age (top-2) reduced accuracy by $\pm 6\%$ and high-risk recall by 3%, but the model remained clinically usable ($\geq 80\%$ sensitivity).

- Removing SystolicBP (BS-only) caused a critical drop in safety: 31.6% of high-risk pregnancies were missed, unacceptable for screening.

The smallest feature subset that achieves the highest classification quality while ensuring safe high-risk detection is:

Three features: Blood Sugar (BS), Systolic Blood Pressure (SBP), and Age

- Accuracy: 75.94%
- High-risk recall: 83.5% (only 1.8% below full model)
- Tree size reduced by 125 nodes (32% smaller than full model)
- Uses only routinely available antenatal data.

This subset is optimal under the InfoGain ranking, balancing performance, interpretability, and clinical safety.

5. Assessment of Ranking Stability and Key Feature Identification

We investigated the stability of feature rankings by comparing two widely used, methodologically distinct approaches:

- Information Gain (InfoGain): measures marginal association between each feature and the class
- ReliefF: estimates feature relevance by considering nearest neighbors and class boundaries (more sensitive to feature interactions)

Rankings Recap (from WEKA outputs):

Rank	InfoGain Score	Feature	ReliefF Score	Feature
1	0.4226	BS	0.0759	BS
2	0.3441	SystolicBP	0.057	SystolicBP
3	0.2408	Age	0.0455	DiastolicBP
4	0.1888	DiastolicBP	0.0454	Age
5	0.0948	HeartRate	0.0213	HeartRate
6	0.032	BodyTemp	0.0187	BodyTemp

5.1. Stability Analysis

- Top-2 features are identical: BS and SystolicBP rank #1 and #2 in both methods, indicating very high stability.
- Age vs DiastolicBP swap at rank 3/4: InfoGain favors Age and DiastolicBP (0.2408 & 0.1888), while ReliefF favors Age and DiastolicBP (0.0455 & 0.0454).
- The difference is negligible:
 - InfoGain $\Delta = 0.2408 - 0.1888 = 0.0718$. It means that age scores 0.0718 higher than DiastolicBP, a moderate gap. So InfoGain clearly prefers Age over DBP.

- ReliefF $\Delta = 0.0455$ & $0.0454 = 0.0001$. It means that DiastolicBP beats Age by only 0.0001 — that's like winning a 100-meter race by 1 millimeter. Essentially a tie.
- This suggests both are moderately relevant, but neither strongly dominates the other.
- HeartRate and BodyTemp consistently lowest, they are both methods agree they contribute minimal discriminative power.

5.2. Quantitative stability (Spearman's rank correlation)

We'll use Spearman's rank correlation coefficient (ρ), a non-parametric measure of monotonic association between two ranked variables. It's perfect for comparing feature rankings.

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Where:

$d_i = \text{Rank}_a(i) - \text{Rank}_b(i)$ = difference in rank for feature i between method A and B

$\sum d_i^2$ = sum of squared rank differences

n = number of features

Let ranks be:

- InfoGain: [1, 2, 3, 4, 5, 6] → BS, SBP, Age, DBP, HR, BT
- ReliefF: [1, 2, 4, 3, 5, 6] → BS, SBP, DBP, Age, HR, BT

Feature	InfoGain Rank (A)	ReliefF Rank (B)	d = A-B	d ²
BS	1	1	1 - 1 = 0	0 ² = 0
SystolicBP	2	2	2 - 2 = 0	0 ² = 0
Age	3	4	3 - 4 = -1	(-1) ² = 1
DiastolicBP	4	3	4 - 3 = +1	1 ² = 1
HeartRate	5	5	5 - 5 = 0	0 ² = 0
BodyTemp	6	6	6 - 6 = 0	0 ² = 0

$$\sum d_i^2 = 0 + 0 + 1 + 1 + 0 + 0 = 2$$

$$\rho = 1 - \frac{6 \cdot 2}{6(6^2 - 1)} = 1 - \frac{12}{210} = 0.943$$

Interpretation of $\rho = 0.943$ indicates very strong agreement rankings are highly stable across algorithms. We can see on table interpretation of ρ Range below:

ρ Range	Interpretation
0.8 – 1.0	Very strong positive correlation
0.6 – 0.8	Strong
0.4 – 0.6	Moderate
< 0.4	Weak

5.3. Which Features Are Most Important? Clinical & Statistical Synthesis

Feature	Statistical Evidence	Clinical Relevance
BS (Blood Sugar)	• #1 in both rankings • Highest InfoGain (0.4226) • Highest ReliefF (0.0759)	Strong predictor of gestational diabetes and preeclampsia; WHO recommends glucose screening in all pregnancies.
SystolicBP	• #2 in both • InfoGain = 0.3441 (2nd highest) • Key root split in all J48 trees	≥ 140 mmHg SystolicBP is a WHO red flag for hypertensive disorders, leading cause of maternal mortality.
DiastolicBP & Age	• Swap at #3/#4, but tightly ranked • Both improve mid-risk discrimination (Task 4)	• DBP helps differentiate isolated vs. combined hypertension • Age modifies risk (teens & >35 years higher risk), used in FIGO risk scores.
HeartRate & BodyTemp	• Consistently last • Removal improves model simplicity without harming high-risk recall	Not part of standard antenatal risk screening; reflect transient states, not chronic risk.

We have different algorithms generate highly similar rankings for this dataset ($\rho = 0.943$). The top two features! Blood Sugar (BS) and Systolic Blood Pressure (SBP), they are robustly and consistently identified as the most important, both statistically and clinically. This convergence increases confidence in their use for simplified risk prediction models.

The minor disagreement at rank 3–4 (Age vs. DiastolicBP) reflects a real clinical trade-off:

- For screening efficiency, BS + SBP alone achieve $\geq 80\%$ high-risk recall.
- For diagnostic refinement, adding either DBP or Age helps separate mid-risk cases, with DBP having a slight edge in interaction-aware methods (ReliefF).

6. Conclusion

This study evaluated feature selection and classification performance on the Maternal Health Risk dataset using WEKA, with the goal of identifying a minimal, clinically feasible, yet accurate prediction model for antenatal risk stratification.

1. Feature rankings are highly stable: Both InfoGain and ReliefF consistently identified Blood Sugar (BS) and Systolic Blood Pressure (SBP) as the two most important predictors (Spearman's $\rho = 0.943$), aligning with WHO and FIGO clinical guidelines.
2. A reduced 3-feature model is optimal under InfoGain ranking:

Using BS, SystolicBP, and Age, J48 achieved 75.93% accuracy and crucially, 83.5% high-risk recall, only 1.8% below the full 6-feature model, while reducing tree complexity by 32%. Removing any of these three (e.g., dropping Age or SBP) led to unacceptable drops in sensitivity (<80%), compromising patient safety.

3. Low-value features can be safely eliminated: HeartRate and BodyTemp ranked lowest in both methods and contributed little to classification. Their removal simplifies data collection without harming high-risk detection — important for resource-limited settings.
4. Clinical deployability: The recommended 3-feature model requires only:
 - a glucometer (BS),
 - a sphygmomanometer (SBP), and
 - patient age, all routinely available even in rural antenatal clinics. The resulting decision tree (125 nodes) is interpretable and suitable for integration into mobile health tools or printed job aids.

In overview, feature selection not only simplifies the model but enhances its practical utility, without sacrificing critical safety performance. This supports the adoption of lightweight, evidence-based screening tools to reduce maternal mortality, especially where advanced diagnostics are unavailable.